



CyClean™

Non-Contact Web Cleaning

The Meech CyClean has been designed in response to increased demand for a compact, high performance, non-contact web cleaning system. Through the application of advanced computational fluid dynamics, Meech has optimised the cleaning efficiency of CyClean to remove and extract contamination to below 1 micron.

A CyClean system comprises three main components; the cleaning head, active static control and an Air Handling Unit (AHUv3).

The double-sided non-contact cleaning head has been designed to provide excellent contamination removal, whilst minimising any risk of web recontamination. On each side of the web there is a powerful blowing airflow that strips the boundary layer and contamination from the web surface. Two vacuum airflows catch all the turbulent air and contamination from within the cleaning head. Each CyClean head incorporates four shockless ionising bars; two positioned on either side of the web, one on entry to and one on exit from the head. Ionising bars ensure neutralisation of any static charges on the web that are holding contaminants to the web surface. The risk of re-attraction of contaminants to the cleaned web surfaces is also minimised.

The AHUv3 provides the airflow and filtration for CyClean.

Applications

CyClean can be used in a wide variety of applications and as it is non-contact, it can be used with all web materials.

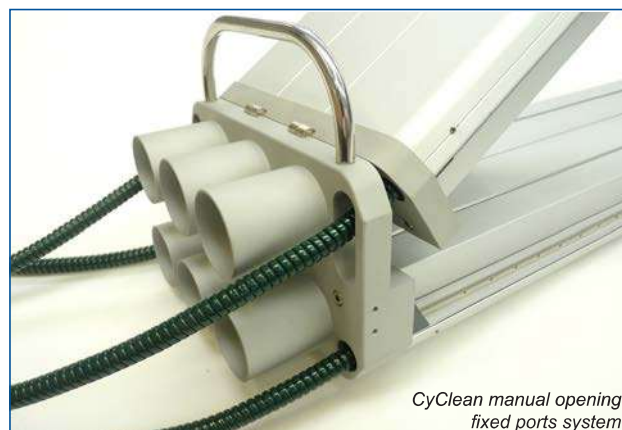
Typical applications include:

- Printing; flexo, digital, gravure and ink jet
- Coating
- Laminating
- Food and medical packaging
- Specialty film processing
- Solar panel substrates
- Battery - aluminium foil
- Blister packs

Installation

The CyClean head unit can be installed at any angle without jeopardising cleaning performance. However, the web must be under constant tension as it passes through the CyClean head.

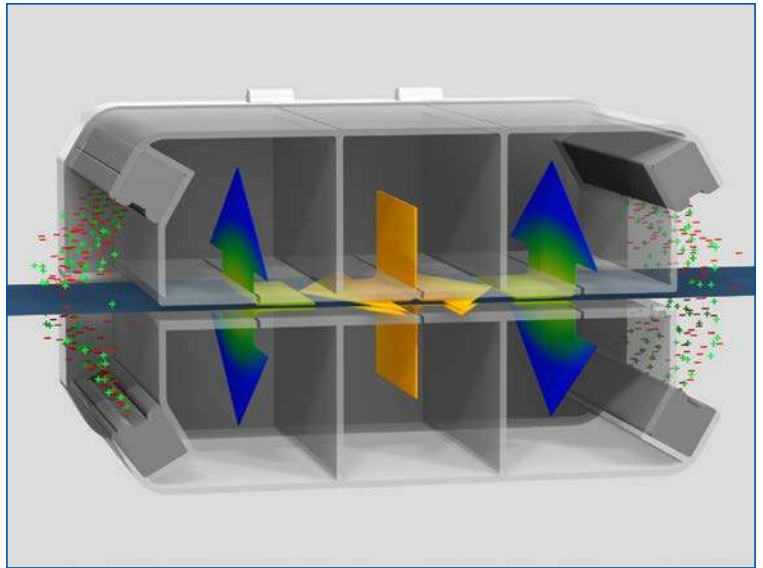
For easier installation, manual opening CyClean Systems feature a unique 'fixed ports' system that allows the ducting connections to remain stationary when the head unit is opened. This makes installing ductwork through a machine back plate far easier.



*CyClean manual opening
fixed ports system*

How CyClean Works

- The web passes through a cloud of ionisation on entry to the CyClean head, which neutralises any static charges.
- The web is then subject to a force of turbulent air created by the blowing and vacuum airflows within the head unit.
- High frequency micro-movements of the web are created by the turbulent air.
- The combination of the high frequency micro-movements and turbulent air shatters the web boundary air layer, releasing the contamination into the vacuum airflows.



- As the web exits the CyClean head, it passes through a second cloud of ionisation to prevent re-contamination of the web.

Design Features

The CyClean is constructed from anodised aluminium. It is a highly rigid and compact system. Based on our unrivalled experience and an extensive development program, we have included several unique features in the system:

Feature	Benefit
Non-contact cleaning	Eliminates risk of surface marking and damage. No interference with web tension or web tracking.
Compact design	Flexible positioning & easy installation.
Integrated static control	Neutralises static charges to optimise cleaning performance and prevent re-contamination.
Unique 'fixed ports' system	The ducting connections to the cleaning head remain stationary, resulting in easier installation.
Air balancing	Allows complete control of the bias between blowing and vacuum airflows.
Automatic vacuum pressure control	Reduces the frequency of required operator adjustments. Maintains optimum cleaning performance automatically.
Hinged, clam shell design	Allows unit to open for easy threading of the web.
No consumable items	Lower ongoing costs.
Quiet operation	Can be installed in enclosed work areas. No H&S issues due to noise.

CyClean Head Opening Configurations

CyClean systems for web widths up to 600mm are supplied as standard in manual opening configuration. If required this can be upgraded to pneumatic opening. Systems for widths of 601mm or greater are supplied as standard with pneumatic opening. Systems to be installed on a vertical web include guided pneumatic cylinders.

"We at Webtech were looking for a web cleaning solution on our new Gidue machine, without the need for constantly changing adhesive rollers. We chose the CyClean non contact system from Meech for its robust design and excellent cleaning abilities. We are delighted with the system performance and look forward to installing more of these in the time to come. The sales and service support from their local representatives has also been a very good."

Managing Director, **Webtech Labels**



CyClean-R™

Non-Contact Web Cleaning

CyClean-R is available as a single-sided cleaning system, with double-sided cleaning configuration available as an option. The CyClean-R is suitable for high-performance contamination removal from low-tension web applications. It utilises the proven CyClean cleaning philosophy of advanced computational fluid dynamics and bypasses the challenge of low tension webs by cleaning the web on a roller.

The system incorporates powerful positive and negative airflows which break the boundary layer of air on the web, efficiently releasing and removing contamination, leading to a clean surface. The airflows of the CyClean-R are controlled to optimise the cleaning performance.

CyClean-R is supplied with the latest Hyperion™ 924IPS static control bars, to maximise the cleaning performance.

Applications

CyClean-R can be used within a wide variety of applications and as a non-contact system, it is suitable for use with all web materials.

Typical applications include:

- Printing; flexo, digital, gravure and ink jet
- Coating
- Laminating
- Food and medical packaging
- Specialty film processing
- Solar panel substrates
- Battery - aluminium foil
- Blister packs

Installation

The CyClean-R head unit is installed over a guide roller. It may be positioned at any angle to best suit the installation, whilst still delivering uncompromised surface cleaning. The CyClean-R overcomes tension issues, as it is designed to be positioned over a roller, where the tension in the web is naturally at it's highest.

How CyClean-R Works

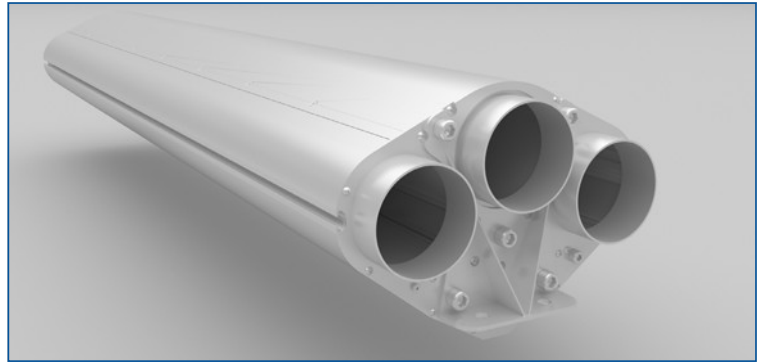
- The web passes through a cloud of ionisation on entry to the CyClean-R head, which neutralises any static charges on the web.
- The web is then subject to a force of turbulent air created by the blowing and vacuum airflows within the head unit.
- Due to the CyClean-R special head design and positioning, the high velocity positive and negative airflows shatter the web boundary air layer, releasing the contamination into the vacuum airflows.
- As the web exits the CyClean head, it passes through a second cloud of ionisation to prevent re-contamination of the web.



Design Features

The CyClean-R has been engineered to provide exceptional results for low-tension applications across a range of industries.

A range of opening options to suit narrow, mid and wide web applications have been developed to provide an ideal solution, considering a number of factors, including ducting access and space constraints. Our range of system configurations mean that the CyClean-R can be flexible to a number of installations, where other web cleaners may not be suitable.



At larger widths of over 1300mm, the CyClean-R includes a top manifold, the 'carapace'. This unique design creates a robust structure that delivers enhanced support for applications up to 9000mm. In addition, the carapace also minimises the use of ducting, leading to a streamlined design.

Feature	Benefit
Non-contact cleaning	Eliminates risk of surface marking and damage. No interference with web tension or web tracking.
Compact design	Space saving design means CyClean-R can be installed on most machines.
Positioning: cleaning over a guide roller	Avoids web tension challenge and ensures effective non-contact cleaning without the danger of damaging the web.
Unique Shape	The air is forced into a certain direction at extremely high velocity, which enhances its cleaning performance.
Robust design	Can clean up to 9000mm web widths.
Air balancing	Allows complete control of the bias between blowing and vacuum airflows.
Automatic vacuum pressure control	Reduces the frequency of required operator adjustments. Maintains optimum cleaning performance automatically.
No consumable items	Lower ongoing costs.
Quiet operation	Can be installed in enclosed work areas. No H&S issues due to noise.

CyClean-R Configurations

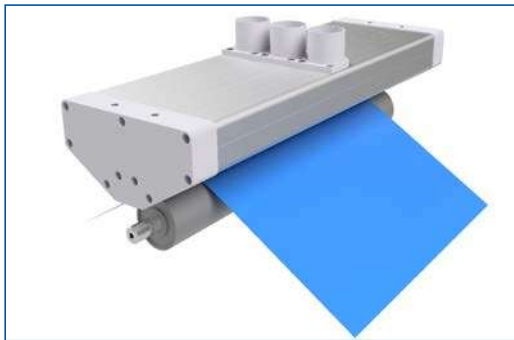
The table below outlines the configurations available, including the opening types, ducting, static control and the AHU required for each. The opening configurations can be seen on page 12-13.

	Web Width (mm)	Opening			Ducting	Qty of 924IPS Bars	Air Handling Units (AHU)
		Fixed	Manual	Pneumatic			
			Option	Option			
Narrow Web	200mm - 699mm	✓	✓	✓	End or Top feed	2	AHUv3 1.1
Mid Web	700mm - 1299mm	✓		✓	2x End feed or 2x Top feed	2	AHUv3 2.2
Wide Web	1300mm - 2399mm	✓		✓	End feed - added carapace	2	AHUv3 3.3
	2400mm - 3999mm	✓		✓	End feed - added carapace	2	AHUv3 6.6
	4000+mm	✓		Special design	End feed - added carapace	4	AHUv3 6.6&3.3

CyClean-R™ System Configurations

In order to best suit your application, we have developed a range of different CyClean-R configurations, considering web width, opening method and ducting connection.

CyClean-R for Narrow Web



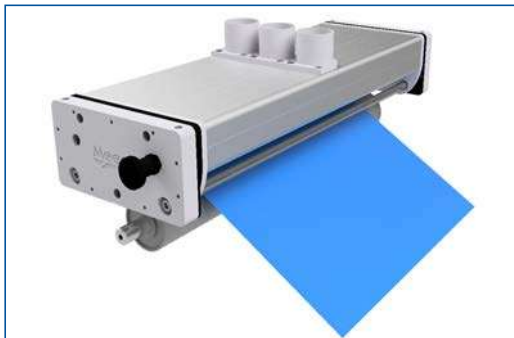
Narrow Web - Fixed: Specification

System Design	Fixed
Ducting Feed	Available as End Feed or TAB Feed
Width Range	200mm - 699mm
Opening range	N/A



Narrow Web - Manual: Specification

System Design	Manual Lift Opening
Ducting Feed	End Feed
Width Range	200mm - 699mm
Opening range	Maximum 90° from closed position



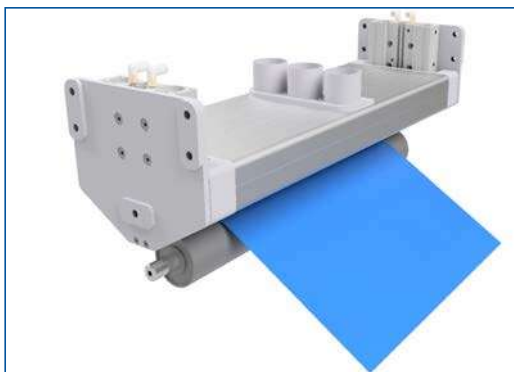
Narrow Web - Manual: Specification

System Design	Manual Tilt Opening
Ducting Feed	TAB Feed
Width Range	200mm - 699mm
Opening range	Maximum 40mm from closed position



Narrow Web - Pneumatic: Specification

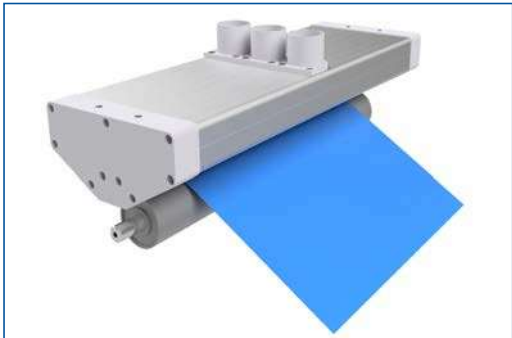
System Design	Pneumatic
Ducting Feed	End Feed
Width Range	200mm - 699mm
Opening range	Maximum 30mm from closed position



Narrow Web - Pneumatic: Specification

System Design	Pneumatic
Ducting Feed	TAB Feed
Width Range	200mm - 699mm
Opening range	Maximum 30mm from closed position

CyClean-R for Mid Web



Mid Web - Fixed: Specification

System Design	Fixed
Ducting Feed	Available as 2x End Feed or 2x TAB Feed
Width Range	700mm -1299mm
Opening range	N/A



Mid Web - Pneumatic: Specification

System Design	Pneumatic
Ducting Feed	End Feed
Width Range	700mm - 1299mm
Opening range	Maximum 30mm from closed position



Mid Web - Pneumatic: Specification

System Design	Pneumatic
Ducting Feed	TAB Feed
Width Range	700mm - 1299mm
Opening range	Maximum 30mm from closed position

CyClean-R for Wide Web



Wide Web - Fixed: Specification

System Design	Fixed with carapace, for enhanced rigidity at larger widths
Ducting Feed	End Feed
Width Range	1300mm - 4000mm
Opening range	N/A



Wide Web - Pneumatic: Specification

System Design	Pneumatic tilt opening with carapace, for enhanced rigidity at larger widths
Ducting Feed	End Feed
Width Range	1300mm - 4000mm
Opening range	18mm